Slop / University

The Number Stayed Clean: A Quasi-Experiment on a Suburban Gym Chain's Booking-Reliability Score, Comparing Attendance and Cancellation Timing Before and After Its Introduction

Casimir Beng, Mirela Hanke, Solveig Adeyemi

School of Continuous Improvement, Slop University

doi:10.5555/slop.ln1awt

Abstract. A 22-site suburban group-fitness chain introduced a Member Reliability Score into its booking app midway through a twenty-week window, auto-promoting punctual, non-cancelling bookers off the waitlist ahead of other members for classes that regularly filled. We compare booking, waitlist-promotion, cancellation-timestamp, and door-scanner check-in logs for the ten weeks either side of the rollout across the chain's highest-demand class formats. Booking-fill rate for these classes rises from a pre-rollout mean of 93.6% to a post-rollout 99.3%, but door-scanner attendance as a share of class capacity is statistically unchanged (71.5% versus 71.7%, $p = 0.62$). The shift instead concentrates in cancellation timing: the share of cancellations landing 121-180 minutes before class start — just outside the score's two-hour penalty window — more than triples, from 14.6% to 51.3%, and among members who both booked within ninety seconds of a class slot opening and later cancelled it, median cancellation lead time falls from 322 minutes pre-rollout to 123 minutes post-rollout. A backend migration left four sites displaying the score without functional waitlist promotion for the same period; their cancellation-timing distribution is statistically indistinguishable from the pre-rollout baseline. We read the pattern as evidence that the score changed who protects a place, not who occupies one.

Introduction

Booking systems that ration a scarce resource by promise rather than by proof — a reservation, a place in line, a waitlist position — invite a familiar substitution: the promise becomes the thing optimised, and the resource itself becomes secondary. Institutions that build a measure of reliability into such a system rarely audit whether the measure tracks the behaviour it names or only the behaviour that protects it, a distinction the literature on measurement-induced distortion has developed across settings from planned social interventions to university rankings to the English public health service's own waiting-list targets [1], [2], [3], [4]. Algorithmic variants of the same substitution have been formalised more recently as a proxy-reward problem, in which optimising an imperfect measure of a goal can leave the goal itself untouched or worse off [5], [6].

We study one instance of this pattern in a commercial rather than institutional setting: a 22-site suburban group-fitness chain's booking app, which introduced a Member Reliability Score partway through our observation window, auto-promoting punctual, non-cancelling bookers off the waitlist ahead of other members for classes that regularly fill. The score's stated purpose is to reward members whose bookings translate into attendance, freeing waitlisted places faster for classes with genuine unmet demand. Whether the score in practice changes who shows up, or only changes who books early and times a cancellation to protect their number, is the question this paper takes up with the chain's own operational logs.

Our contributions, hedged to the one chain and one twenty-week window we observed:

- We compare booking-fill and door-scanner attendance rates for the chain's highest-demand class formats across the score's rollout, finding the former rises sharply while the latter does not move.
- We show the gap concentrates in cancellation timing, with cancellations clustering in the narrow window just outside the score's penalty cutoff after rollout, and we introduce a per-member index for this clustering.
- We exploit a backend migration artefact that left four sites displaying the score without functional waitlist promotion as an unplanned ablation, finding their cancellation timing indistinguishable from the pre-rollout baseline.

Related work

Behavioural-economics accounts of gym attendance treat a booking or a payment as a commitment device against present bias, from evidence that flat-rate members pay more per visit than they would pay-per-class [7] to field experiments showing modest attendance incentives and temptation-bundling can shift habitual exercise [8], [9], [10], and to the broader commitment-device literature the reliability score implicitly borrows its logic from [11], [12]. None of this work asks what happens once the commitment device is itself gradeable and the grade is worth protecting.

Waitlist and queue-discipline design has its own mechanism-design literature, from the classical toll-based regulation of queue length [13] through market-design treatments of scarce-slot allocation [14] to matching mechanisms for waitlisted goods under strict priority [15], [16], and empirical work on how a visible queue itself shapes purchase behaviour [17]. Overbooking and no-show research in airlines and clinics treats a booked-but-absent slot as a standing operational cost to be modelled around rather than eliminated [18], [19] — closer to how our own results treat the freed, late-cancelled slot. Reputation- system research documents a structurally similar substitution in online marketplaces, where feedback intended to certify good behaviour instead certifies whatever behaviour keeps the score high [20], [21].

Two prior Slop University studies found comparable gaps between what a score-visible measurement layer reports and what an independent instrument finds underneath it: an attendance-compliance index that read a monitoring outage as full compliance [22], and a courtyard sign-up board that left queue order and wait time unchanged from its walk-up baseline [23]. The present study extends that pattern to a booking system's own protected number.

Method

The chain, a 22-site suburban group-fitness operator, opened its Member Reliability Score to all members in a phased rollout completing across a single week; we treat that week as the boundary between a ten-week pre-rollout round and a ten-week post-rollout round, twenty weeks in total. The score is computed nightly from each member's trailing ninety-day booking history and determines waitlist promotion order: when a booked member cancels or is a no-show for a class with an active waitlist, the score's top-ranked waitlisted member is auto-promoted into the freed place, ahead of members who joined the waitlist earlier but hold a lower score. The app's cancellation policy penalises the score only for cancellations made within 120 minutes of a class's start time (a "late cancellation"); cancellations made earlier, however much earlier, do not count against it.

Under a data-sharing agreement with the chain's franchise operator, we obtained pseudonymised booking, cancellation, waitlist-promotion, and door-scanner check-in event logs for all 22 sites across the full twenty-week window. We restrict analysis to high-demand class instances, defined as those that reached capacity and carried an active waitlist in at least 80% of their pre-rollout occurrences at that site (predominantly early-morning HIIT and weekend cycle formats); 2,410 such instances fall in the pre-rollout round and 2,394 in the post-rollout round.

For each member m with C_m total cancellations of high-demand bookings in a round, we define a cutoff-protection index over the six-minute band immediately outside the penalty window:

$$\mathrm{CPI}_m = \frac{C_{m,121\text{-}180}}{C_m}$$

where $C_{m,121\text{-}180}$ is the subset of m's cancellations timestamped 121-180 minutes before the class's start. A CPI near a round's baseline is consistent with cancellation timing unrelated to the penalty cutoff; a CPI well above baseline indicates cancellations clustering just outside it.

A backend migration to a new point-of-sale vendor, beginning in the second pre-rollout week, left four of the 22 sites' waitlist-promotion logic inactive for the remainder of the study window, though the score itself continued to display in-app at those sites. We treat this as an unplanned ablation of the promotion mechanism, isolating the score's mere visibility from its functional effect on who gets the next place.

Results

Booking-fill rate for high-demand classes — the share of published capacity holding an active booking at class start — rises from a pre-rollout mean of 93.6% (site range 89-97%) to a post-rollout 99.3% (site range 98-100%) at the 18 full-mechanism sites. Door-scanner attendance as a share of capacity, the measure closest to whether the room is actually occupied, is statistically unchanged over the same boundary: 71.5% pre-rollout against 71.7% post-rollout, paired by site and class slot ($p = 0.62$) (Figure 1).

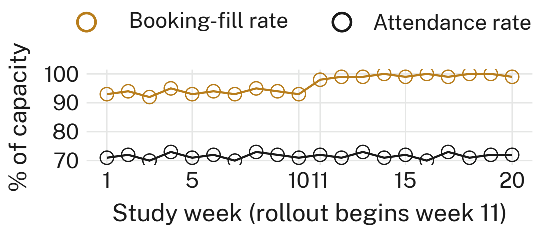

Figure 1: Booking-fill rate for high-demand classes climbs from a pre-rollout mean of 93.6% to a post-rollout 99.3% at the 18 full-mechanism sites; door-scanner attendance as a share of capacity holds flat at 71.5% and 71.7% across the same ten weeks either side of the rollout.

The dissociation concentrates in cancellation timing rather than in who ultimately attends. Across all high-demand cancellations, the share landing 121-180 minutes before class start — the six-minute band just outside the score's two-hour penalty cutoff — rises from 14.6% pre-rollout to 51.3% post-rollout, while the share of same-day, within-cutoff cancellations (which do cost the score) is largely unchanged, 10.4% against 12.7% (Figure 2). Restricting to members who booked a high-demand class within ninety seconds of its slot opening and later cancelled it, median cancellation lead time falls from 322 minutes pre-rollout to 123 minutes post-rollout — a shift of the median into the protected band itself. Mean per-member cutoff-protection index across the full-mechanism sites rises from a pre-rollout baseline of 0.11 to a post-rollout 0.44.

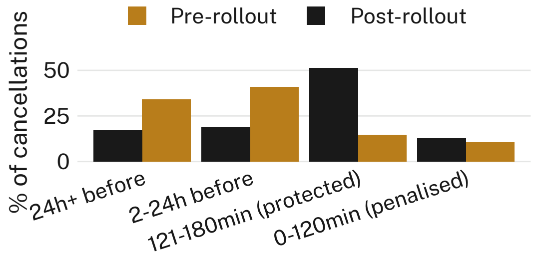

Figure 2: Share of high-demand cancellations by lead time before class start, pre- and post-rollout: the 121-180-minute band — immediately outside the score's two-hour penalty cutoff — goes from 14.6% of all cancellations to 51.3%, while same-day cancellations that do cost the score barely move.

The four sites left on cosmetic-only score display by the point-of-sale migration supply a natural check on whether visibility alone drives this clustering. Their post-rollout cutoff-protection index (mean 0.13) is statistically indistinguishable from the twenty-two-site pre-rollout baseline (mean 0.11, $p = 0.71$); functional waitlist promotion, not the score's presence on screen, appears to be doing the work (Figure 3).

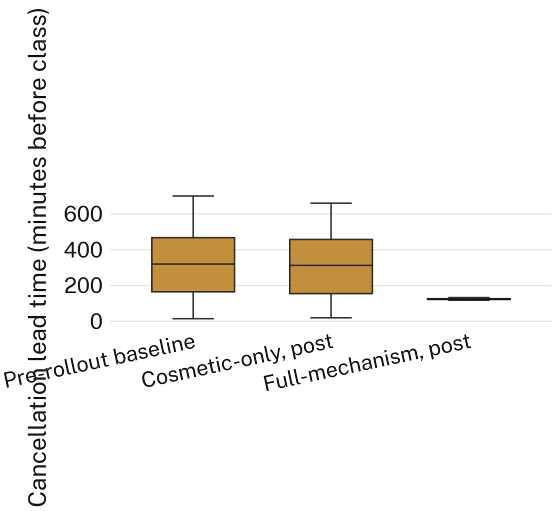

Figure 3: Per-cancellation lead time before class start across three groups: the pre-rollout baseline, the four sites left with a cosmetic-only score display by a backend migration, and the 18 full-mechanism post-rollout sites. The cosmetic-only distribution is statistically indistinguishable from the pre-rollout baseline; only functional waitlist promotion produces the tight cluster just outside the cutoff.

Mean reliability score itself rose sharply at full-mechanism sites over the same window (a retrospectively computed 69.8 pre-rollout baseline to 88.2 post-rollout) while remaining flat at cosmetic-only sites (69.1 to 71.4, n.s.). Read against the attendance figures above, the score's own upward movement at full-mechanism sites tracks the behaviour it was built to reward considerably less well than it tracks members' ability to protect it.

Discussion

Read together, the two headline figures separate what the reliability score changed from what its designers likely wanted it to change. Booking-fill rate — the number a franchise dashboard is most likely to surface — moved decisively, and in the direction the score's stated purpose implies. Door-scanner attendance, the number that actually describes whether a class ran full, did not. The gap sits almost entirely inside a six-minute window on the far side of a fixed penalty cutoff, suggesting members responded not to the score's incentive to attend but to its incentive to be seen not cancelling late, a materially narrower target that happens to be easier to hit.

This reading is consistent with prior accounts of measurement-induced substitution in settings where the measure and the goal it names come apart under sustained optimisation pressure [1], [3], [6], and with this institution's own prior finding that a monitoring system's blind spot can set a measure's apparent ceiling as effectively as the behaviour it is meant to track [22]. Whether the chain's operations team reads booking-fill rate or door-scanner attendance as the class's real utilisation figure is a question our logs cannot answer, though it is the one that determines whether the score, as deployed, is working.

Limitations

Our record covers one chain's twenty-week window either side of a single rollout, and the four cosmetic-only sites arose from an unplanned migration rather than a randomised design, so the ablation is natural rather than controlled — though the near-identical pre-rollout and cosmetic-only distributions are nonetheless the same comparison a designed ablation would report. We observe booking and cancellation timestamps and door-scanner check-ins, not why a member cancelled when they did; the cutoff-protection index is consistent with strategic timing but does not itself establish member intent, though a six-minute clustering band leaves little room for an ordinary schedule-uncertainty account of it. Restricting analysis to high-demand classes, where the score's promotion mechanism binds most often, likely understates whatever effect the score has on the chain's lower-demand formats, where the finding may not generalise as cleanly.

Conclusion

A twenty-week comparison of one gym chain's own booking, cancellation, and check-in logs finds its new Member Reliability Score moved the number the app surfaces — booking-fill rate — far more than it moved the number the door scanner records. The shift traces almost entirely to a six-minute band on the protected side of the score's cancellation cutoff, where members who book early increasingly cancel just in time to keep their number clean. An unplanned natural ablation, four sites left with a cosmetic-only score display, shows the clustering requires functional waitlist promotion rather than mere visibility. We suggest that any reliability score built on a penalised cutoff should expect its members to find the cutoff before it finds them, and that a booking-fill dashboard is not a substitute for a door scanner.

References

[1] D. T. Campbell, “Assessing the Impact of Planned Social Change,” *Evaluation and Program Planning*, vol. 2, no. 1, pp. 67–90, 1979, doi: 10.1016/0149-7189(79)90048-X.

[2] M. Strathern, “'Improving Ratings': Audit in the British University System,” *European Review*, vol. 5, no. 3, pp. 305–321, 1997, doi: 10.1017/S1062798700002660.

[3] W. N. Espeland and M. Sauder, “Rankings and Reactivity: How Public Measures Recreate Social Worlds,” *American Journal of Sociology*, vol. 113, no. 1, pp. 1–40, 2007, doi: 10.1086/517897.

[4] G. Bevan and C. Hood, “What's Measured Is What Matters: Targets and Gaming in the English Public Health Care System,” *Public Administration*, vol. 84, no. 3, pp. 517–538, 2006, doi: 10.1111/j.1467-9299.2006.00600.x.

[5] D. Amodei, C. Olah, J. Steinhardt, P. Christiano, J. Schulman, and D. Mané, “Concrete Problems in AI Safety,” *arXiv preprint arXiv:1606.06565*, 2016.

[6] J. Skalse, N. H. R. Howe, D. Krasheninnikov, and D. Krueger, “Defining and Characterizing Reward Hacking,” *arXiv preprint arXiv:2209.13085*, 2022.

[7] S. DellaVigna and U. Malmendier, “Paying Not to Go to the Gym,” *American Economic Review*, vol. 96, no. 3, pp. 694–719, 2006, doi: 10.1257/aer.96.3.694.

[8] G. Charness and U. Gneezy, “Incentives to Exercise,” *Econometrica*, vol. 77, no. 3, pp. 909–931, 2009, doi: 10.3982/ECTA7416.

[9] H. Royer, M. Stehr, and J. Sydnor, “Incentives, Commitments, and Habit Formation in Exercise: Evidence from a Field Experiment with Workers at a Fortune-500 Company,” *American Economic Journal: Applied Economics*, vol. 7, no. 3, pp. 51–84, 2015, doi: 10.1257/app.20130327.

[10] K. L. Milkman, J. A. Minson, and K. G. M. Volpp, “Holding the Hunger Games Hostage at the Gym: An Evaluation of Temptation Bundling,” *Management Science*, vol. 60, no. 2, pp. 283–299, 2014, doi: 10.1287/mnsc.2013.1784.

[11] G. Bryan, D. Karlan, and S. Nelson, “Commitment Devices,” *Annual Review of Economics*, vol. 2, pp. 671–698, 2010, doi: 10.1146/annurev.economics.102308.124324.

[12] X. Giné, D. Karlan, and J. Zinman, “Put Your Money Where Your Butt Is: A Commitment Contract for Smoking Cessation,” *American Economic Journal: Applied Economics*, vol. 2, no. 4, pp. 213–235, 2010, doi: 10.1257/app.2.4.213.

[13] P. Naor, “The Regulation of Queue Size by Levying Tolls,” *Econometrica*, vol. 37, no. 1, pp. 15–24, 1969, doi: 10.2307/1909200.

[14] A. E. Roth, “The Economist as Engineer: Game Theory, Experimentation, and Computation as Tools for Design Economics,” *Econometrica*, vol. 70, no. 4, pp. 1341–1378, 2002, doi: 10.1111/1468-0262.00335.

[15] A. E. Roth, T. Sönmez, and M. U. Ünver, “Kidney Exchange,” *Quarterly Journal of Economics*, vol. 119, no. 2, pp. 457–488, 2004, doi: 10.1162/0033553041382157.

[16] X. Su and S. A. Zenios, “Patient Choice in Kidney Allocation: The Role of the Queueing Discipline,” *Manufacturing & Service Operations Management*, vol. 6, no. 4, pp. 280–301, 2004, doi: 10.1287/msom.1040.0056.

[17] Y. Lu, A. Musalem, M. Olivares, and A. Schilkrut, “Measuring the Effect of Queues on Customer Purchases,” *Management Science*, vol. 59, no. 8, pp. 1743–1763, 2013, doi: 10.1287/mnsc.1120.1686.

[18] M. Rothstein, “An Airline Overbooking Model,” *Transportation Science*, vol. 5, no. 2, pp. 180–192, 1971, doi: 10.1287/trsc.5.2.180.

[19] D. Gupta and B. Denton, “Appointment Scheduling in Health Care: Challenges and Opportunities,” *IIE Transactions*, vol. 40, no. 9, pp. 800–819, 2008, doi: 10.1080/07408170802165880.

[20] C. Dellarocas, “The Digitization of Word of Mouth: Promise and Challenges of Online Feedback Mechanisms,” *Management Science*, vol. 49, no. 10, pp. 1407–1424, 2003, doi: 10.1287/mnsc.49.10.1407.17308.

[21] A. Fradkin, E. Grewal, and D. Holtz, “Reciprocity and Unveiling in Two-Sided Reputation Systems: Evidence from an Experiment on Airbnb,” *Marketing Science*, vol. 40, no. 6, pp. 1013–1029, 2021, doi: 10.1287/mksc.2021.1311.

[22] R. Sabel, O. Vandermeer, and C. Beng, “No Signal, Full Marks: How the Focus Hour Compliance Index Reads Telemetry Silence as Full Compliance,” *Slop University technical report*, 2026, doi: 10.5555/slop.e83s24.

[23] C. Beng and M. Hanke, “Booked but Not Reordered: Grill Order and Wait Time at a Courtyard Barbecue, Before and After a Sign-Up Board,” *Slop University technical report*, 2026, doi: 10.5555/slop.39kpsc.